

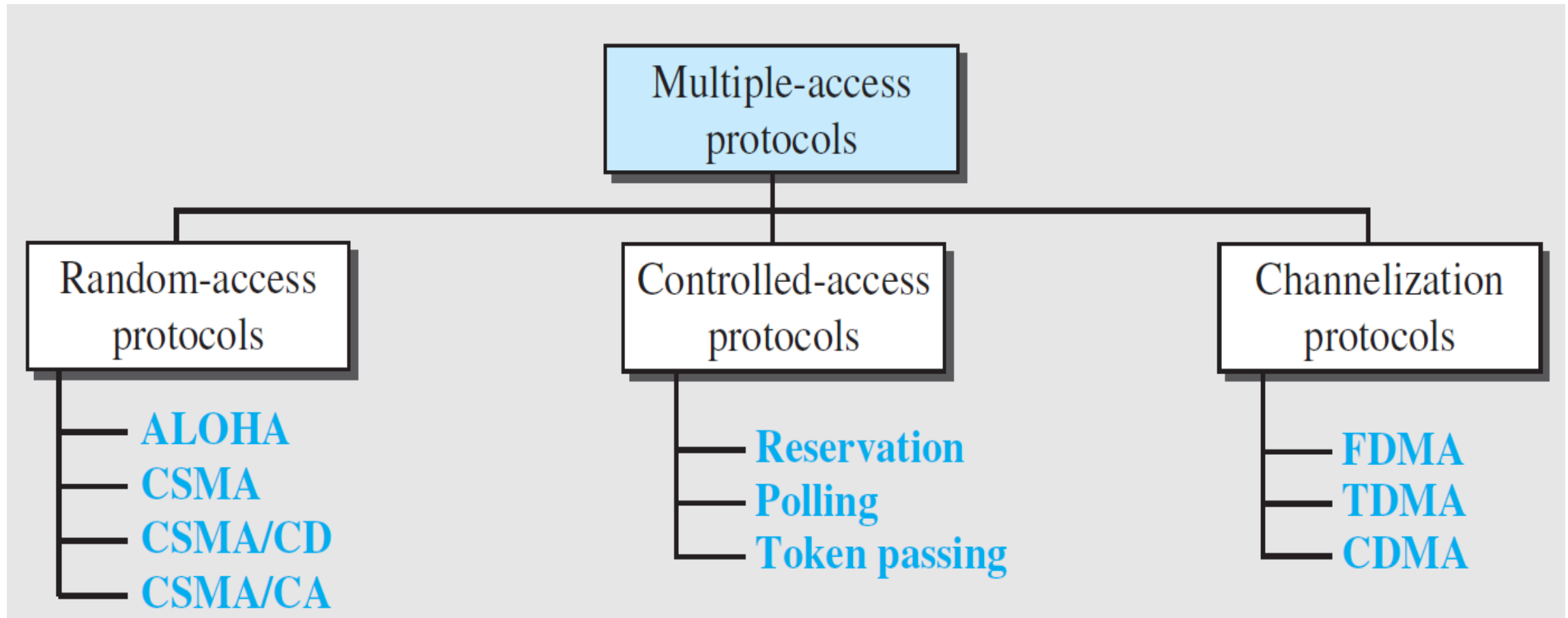
DATA LINK LAYER

Module 2

Channel Access Protocol:

- Channel Access Protocols are rules used in the Data Link Layer to **control how multiple devices share** and **access** a common communication channel.
- When many devices are connected to the **same shared medium**, such as a wireless channel or a shared Ethernet network, **two or more devices** may try to **transmit** at the **same time**. This can cause a **collision**.
- Therefore, Channel Access Protocols **determine how multiple devices** share a **common communication channel** without causing **unnecessary conflicts or collisions**.

Classification of multiple-access protocols



Random Access Protocol:

- **No central control:** All stations have **equal rights** to **access** the shared medium.
- **Random access:** A station can **transmit** whenever it has data, after checking the **medium (idle or busy)**. If the **medium** is **idle**, station can **transmit data, otherwise not**.
- **Contention & collision:** Stations **compete** for the **medium**, so **simultaneous transmissions** may cause **collisions**.
- **Main protocols:** ALOHA, CSMA, CSMA/CD, and CSMA/CA.

ALOHA:

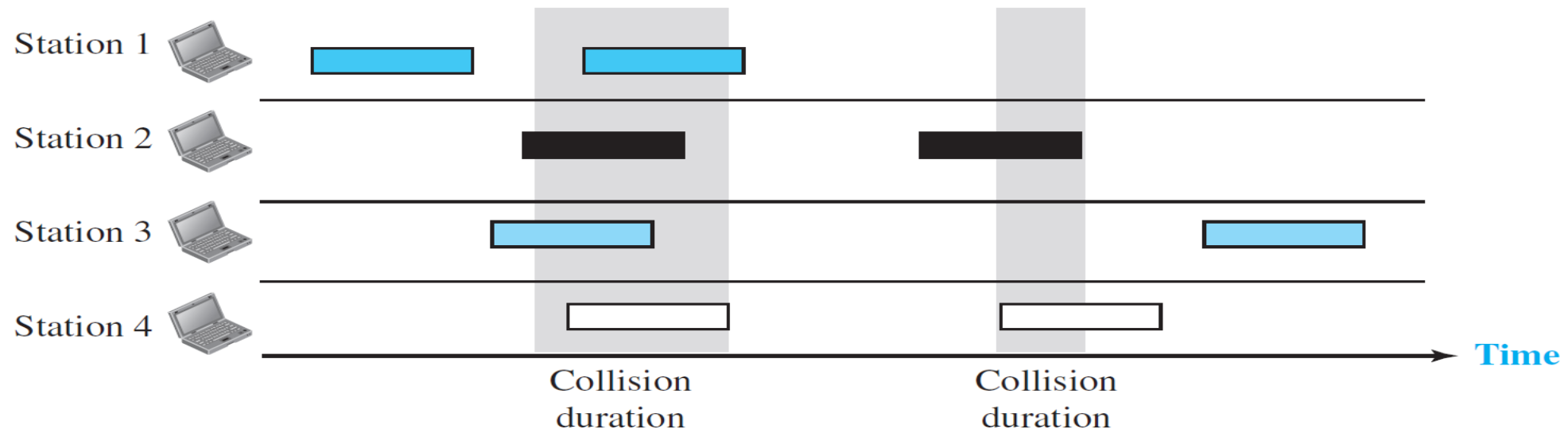
- ALOHA is the **first random-access** method, developed at the **University of Hawaii** in the **early 1970s**. It was **mainly designed** for **wireless communication**.
- In ALOHA, many stations share the **same communication medium**. If two stations send **data** at the **same time**, their **data collide**, and the data becomes **damaged or unreadable**.
- There are two types of ALOHA protocols:
 1. Pure ALOHA
 2. Slotted ALOHA

Pure ALOHA:

- The **original ALOHA** protocol is called pure ALOHA.
- In pure ALOHA, each station **sends a frame** whenever it has a **frame to send** (multiple access).
- However, since there is **only one channel** to share, there is the **possibility** of **collision** between frames from different stations.

Example:

- There are **four stations** that contend with one another for **access to the shared channel**.
- Each station **sends two frames**; there are a total of **eight frames** on the shared medium.
- Some of these frames **collide** because **multiple frames** are in **contention** for the **shared channel**.



Frames in a pure ALOHA network

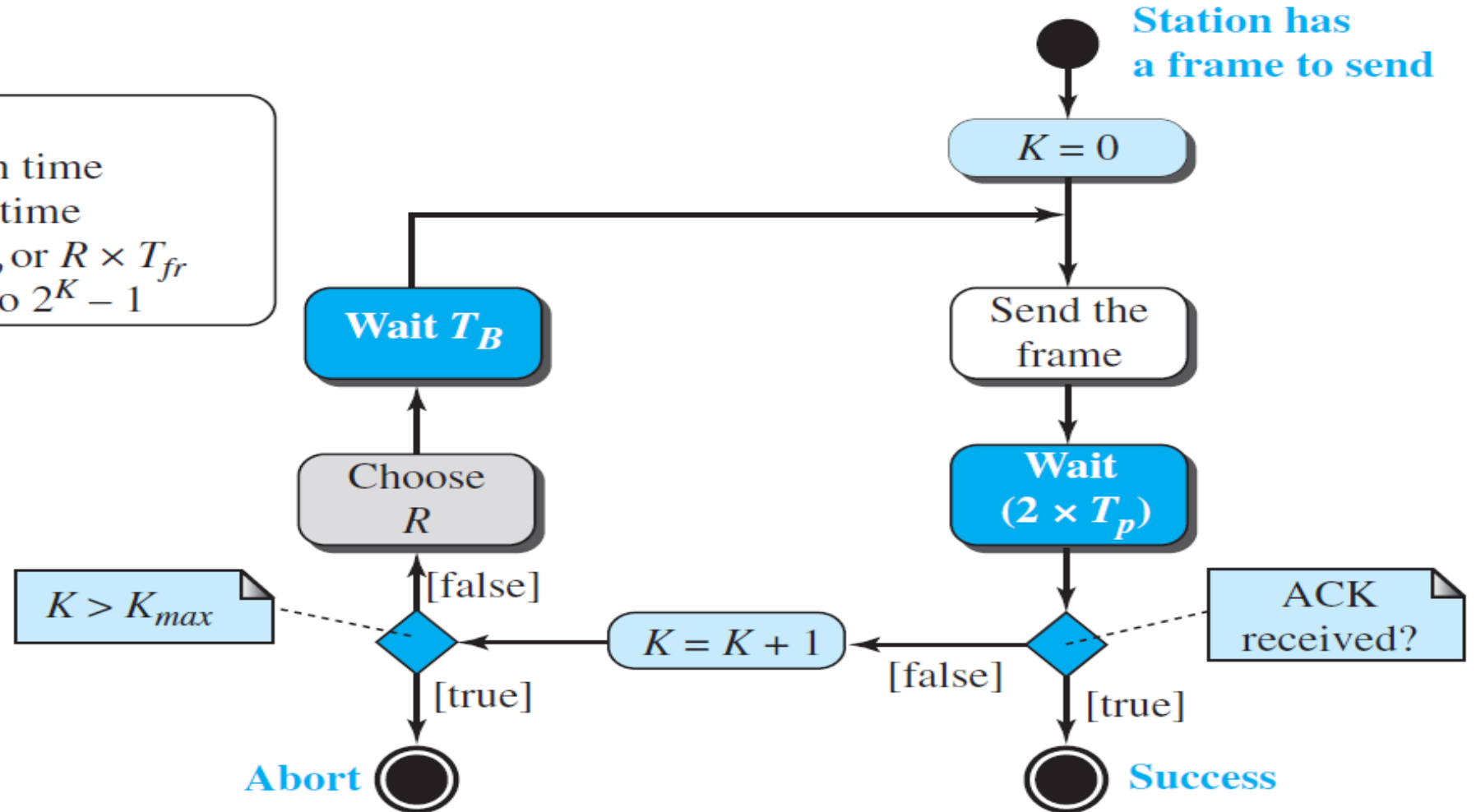
- Only **two frames survive**: one frame from station 1 and one frame from station 3.
- Even if **one bit of a frame** coexists on the channel with **one bit from another frame**, there is a **collision** and both will be **destroyed**.
- We need to **resend the frames** that have been **destroyed** during transmission.

Procedure for pure ALOHA protocol:

- The pure ALOHA protocol **relies on acknowledgments** from the receiver. When a station **sends a frame**, it expects the receiver to **send an acknowledgment**.
- If the acknowledgment **does not arrive** after a **time-out period**, the station assumes that the **frame (or the acknowledgment)** has been **destroyed** and **resends the frame**.
- A **collision** involves **two or more stations**. If all these stations **try to resend** their frames after the **time-out**, the frames will **collide again**.
- Pure ALOHA dictates that when the **time-out period passes**, each station **waits a random amount of time** before resending its frame.
- The randomness will **help avoid more collisions**. This random amount of waiting time is known as **backoff time T_B** .

Legend

K : Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time
 T_B : (Backoff time): $R \times T_p$ or $R \times T_{fr}$
 R : (Random number): 0 to $2^K - 1$



Procedure for pure ALOHA protocol

- The **time-out period** is equal to the **maximum** possible **round-trip propagation delay**, which is **twice** the amount of **time** required **to send a frame** between the two most **widely separated stations** ($2 \times T_p$).
- **Round-trip propagation delay** is the **time** taken for a **signal** to **travel** from the **sender** to the **receiver** and then **back** from the **receiver** to the **sender**.
- The backoff time T_B is a **random value** that normally **depends on K** (the number of **attempted unsuccessful transmissions**).
- For each retransmission, a multiplier $R = 0$ to $2^K - 1$ is **randomly chosen** and **multiplied** by T_p (**maximum propagation time**) or T_{fr} (**the average time required to send out a frame**) to find T_B .

Example: The stations on a wireless ALOHA network are a maximum of 600 km apart. If the signals propagate at 3×10^8 m/s, find Propagation time (T_p).

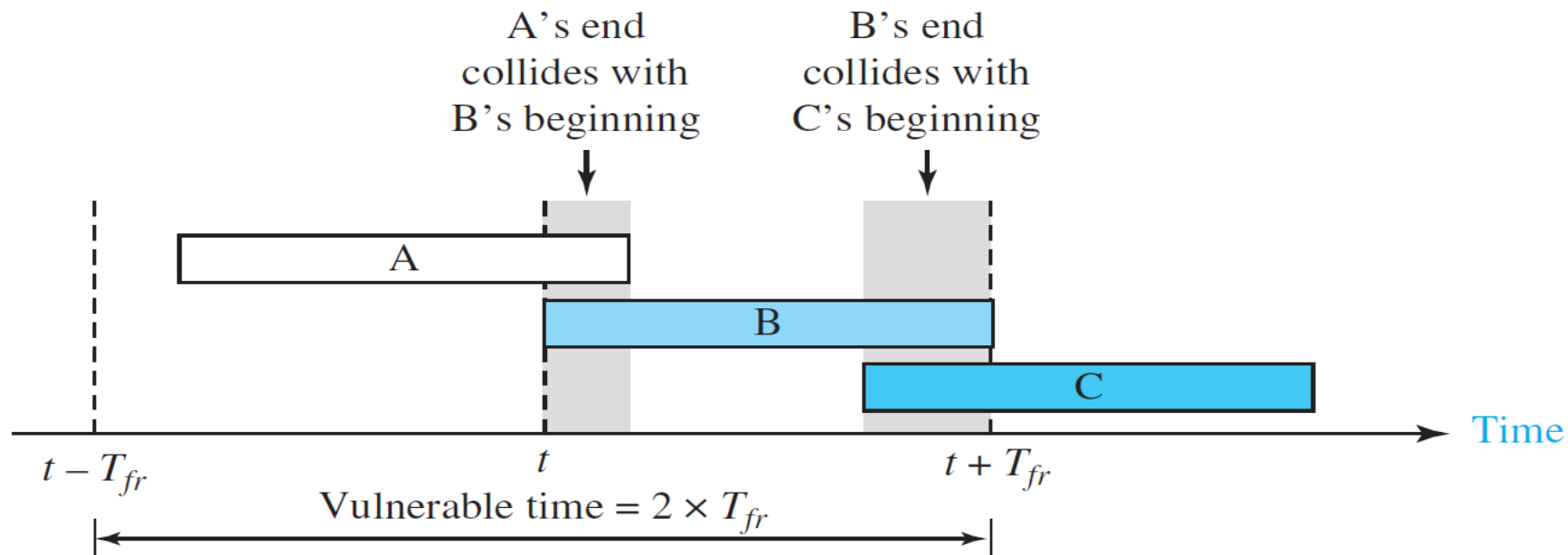
Solution: The distance between stations = $600 \text{ km} = 6 \times 10^5 \text{ m}$

$$\text{Signal Speed} = 3 \times 10^8 \text{ m/s}$$

$$\text{Propagation time } (T_p) = \frac{\text{Distance}}{\text{Signal Speed}} = \frac{6 \times 10^5 \text{ m}}{3 \times 10^8 \text{ m/s}} = 2 \text{ ms}$$

Vulnerable time in pure ALOHA:

- The **length of time** in which there is a **possibility of collision** is known as vulnerable time.
- We assume that the stations send **fixed-length frames** with each frame taking **average transmission time** T_{fr} seconds **to send**.



Vulnerable time for pure ALOHA protocol

- Station **B** starts to send a frame at time t . Now imagine station **A** has started to send its frame after $t - T_{fr}$. This leads to a collision between the frames from station B and station A.
- On the other hand, suppose that station **C** starts to send a frame before time $t + T_{fr}$. Here, there is also a collision between frames from station B and station C.
- The vulnerable time during which a collision may occur in pure ALOHA is 2 times the frame transmission time.

$$\text{Vulnerable Time} = 2 \times T_{fr}$$

Example: A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

Solution: In Pure ALOHA, a frame can collide if another station starts transmitting at any time while the frame is being transmitted.

Given: Frame Size = 200 bits and Data rate = 200 kbps = 200000 bits/sec.

Calculate transmission time: $T_{fr} = \frac{\text{Frame Size}}{\text{Data rate}} = \frac{200 \text{ bits}}{200000 \text{ bits/sec}} = 1 \text{ ms}$

So, the frame takes 1 ms to transmit.

For a Pure ALOHA frame to be collision-free, no other station should start transmitting during the vulnerable period.

The vulnerable period in Pure ALOHA is: $\text{Vulnerable Time} = 2 \times T_{fr} = 2 \times 1 \text{ ms} = 2 \text{ ms}$

Therefore, to make the 200-bit frame collision-free, no other station should transmit during a 2 ms vulnerable period around the frame's transmission.

Throughput in pure ALOHA:

- Throughput is the **average** number of **frames successfully transmitted** during a **given time period**.
- Let **G** is the **average number of frames** generated by the system during **one frame transmission time**. Then the **average number of successfully transmitted frames** for pure ALOHA is $S = G \times e^{-2G}$.
- The maximum throughput, $S_{\max}$, is **0.184** for $G=1/2$. This means that if **one-half** of a **frame** is generated during **one frame transmission time** (i.e., one frame is generated during two frame transmission times), then **18.4% of the frames** reach their **destination successfully**.

Example: A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces 1000 frames per second?

Solution: Frame Size = 200 bits and Data rate = 200 kbps = 200000 bits/sec.

Calculate transmission time: $T_{fr} = \frac{\text{Frame Size}}{\text{Data rate}} = \frac{200 \text{ bits}}{200000 \text{ bits/sec}} = 1 \text{ ms}$

Therefore, in 1 sec, number of frames transmitted = 1000.

in 1 ms. i.e. 10^{-3} sec, number of frame transmitted = $1000 \times 10^{-3} = 1 = G$

The throughput $S = G \times e^{-2G} = e^{-2} = 0.135 = 13.5 \%$. ($e = 2.71828$)

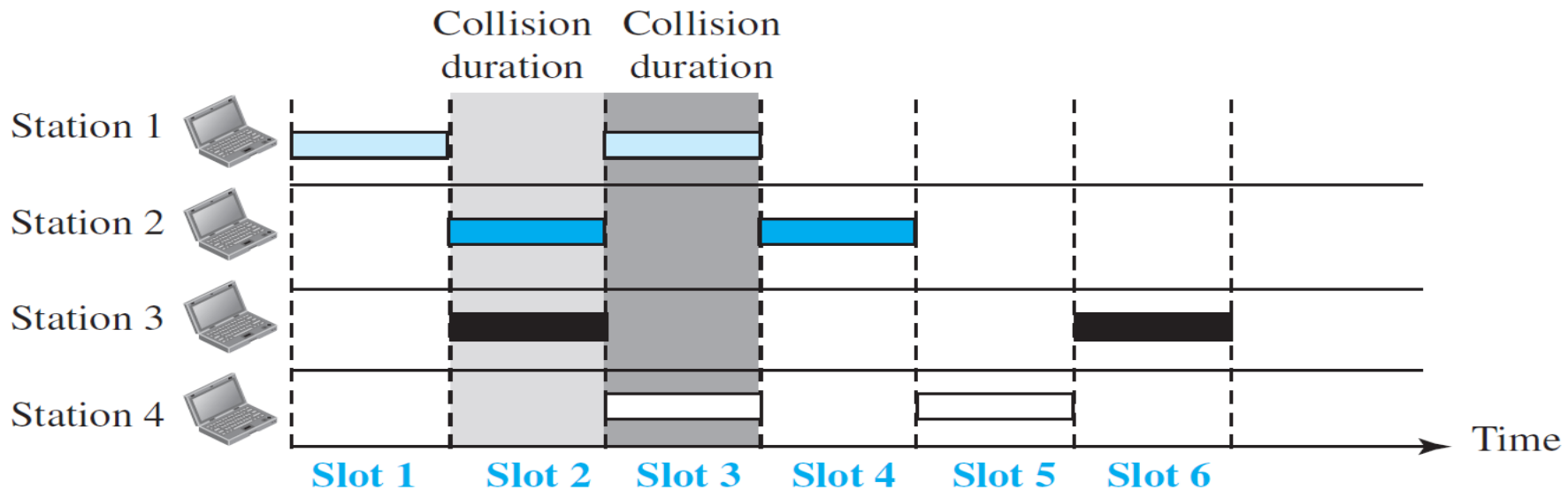
This means that the throughput is $1000 \times 0.135 = 135$ frames.

Only 135 frames out of 1000 will probably survive.

Slotted ALOHA:

- Slotted ALOHA was invented to **improve the efficiency** of pure ALOHA.
- In slotted ALOHA, we divide the **time** into **slots of T_{fr} seconds** and **force** the station to **send only** at the **beginning of the time slot**.

Example:

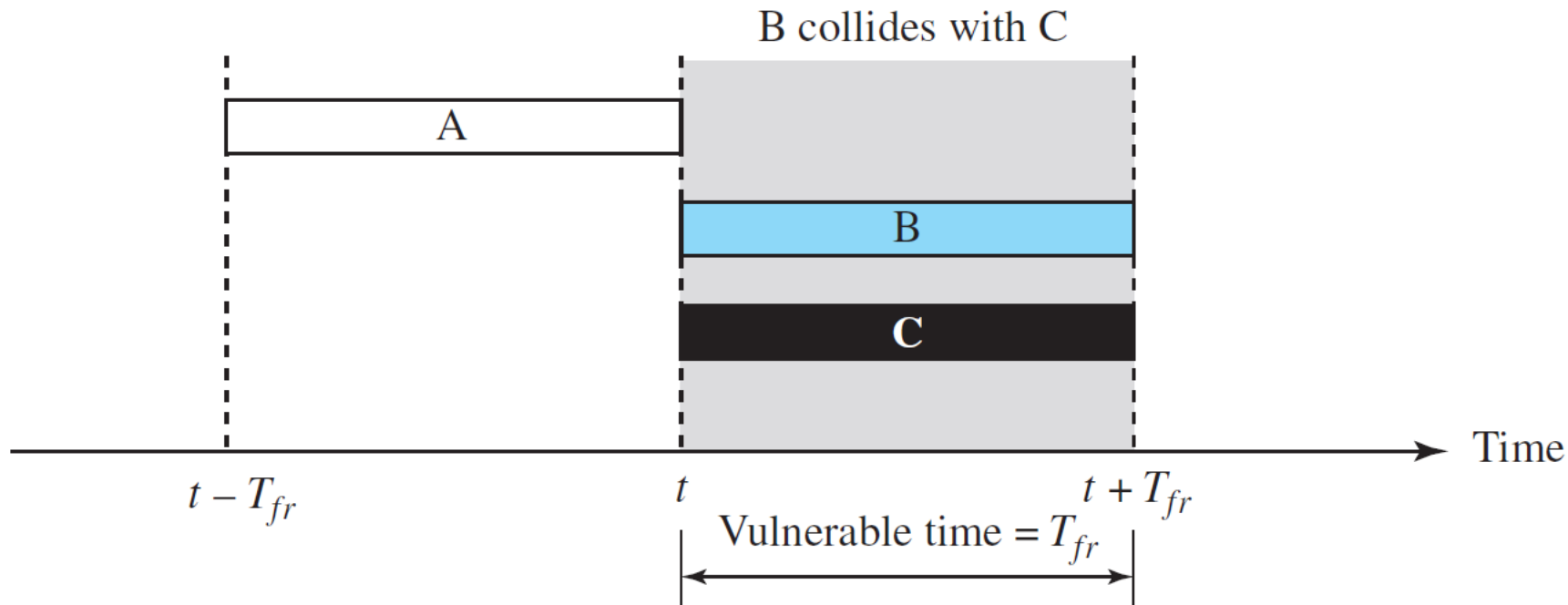


Frames in a slotted ALOHA network

- Because a station is allowed to **send only** at the **beginning** of the **synchronized time slot**, if a **station misses** this moment, it must **wait until** the **beginning** of the **next time slot**.
- This means that the station which **started at the beginning** of this slot has **already finished sending** its frame.

- There is still the **possibility of collision** if two stations try to **send at the beginning** of the **same time slot**. However, the **vulnerable time** is now reduced to one-half, equal to T_{fr} .

$$\text{Vulnerable Time} = T_{fr}$$



Vulnerable time for slotted ALOHA protocol

Throughput in slotted ALOHA:

- The **average** number of **successful transmissions** for slotted ALOHA is $S = G \times e^{-G}$.
- The maximum throughput $S_{\max}$ is **0.368**, when $G = 1$. In other words, if **one frame** is generated during **one frame transmission time**, then **36.8 percent** of these **frames** reach their **destination successfully**.

Example: A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces 1000 frames per second.

Solution: Frame Size = 200 bits and Data rate = 200 kbps = 200000 bits/sec.

Calculate transmission time: $T_{fr} = \frac{\text{Frame Size}}{\text{Data rate}} = \frac{200 \text{ bits}}{200000 \text{ bits/sec}} = 1 \text{ ms}$

Therefore, in 1 sec, number of frames transmitted = 1000.

in 1 ms. i.e. 10^{-3} sec, number of frame transmitted = $1000 \times 10^{-3} = 1 = G$

The throughput $S = G \times e^{-G} = e^{-1} = 0.368 = 36.8 \%$.

This means that the throughput is $1000 \times 0.368 = 368$ frames.

Only 368 frames out of 1000 will probably survive.



THANK YOU